HW 1

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9/29/2026

1

(a) Given random variable X where $a \leq X \leq b$, we can prove that $X - E[X]$ is $\frac{(b-a)^2}{4}$-subgaussian. Let $Y = X - E[X]$ and the log MGF is $\psi(\lambda) = \log E[e^{\lambda Y}]$. Using the identity from the hint, we can bound $\psi''(\lambda)$ as follows:

$$\text{Let } M(\lambda) = E[e^{\lambda Y}]$$
$$\psi(\lambda) = \log M(\lambda)$$
$$\psi'(\lambda) = \frac{M'(\lambda)}{M(\lambda)} = \frac{E[Ye^{\lambda Y}]}{E[e^{\lambda Y}]}$$
$$\psi''(\lambda) = \frac{M''(\lambda)M(\lambda) - (M'(\lambda))^2}{(M(\lambda))^2} = \frac{E[Y^2e^{\lambda Y}]E[e^{\lambda Y}] - (E[Ye^{\lambda Y}])^2}{(E[e^{\lambda Y}])^2}$$

Since $Y = X - E[X]$ and subtracting a constant does not change variance:

$$\psi''(\lambda) = \frac{E[Y^2e^{\lambda Y}]}{E[e^{\lambda Y}]} - \left(\frac{E[Ye^{\lambda Y}]}{E[e^{\lambda Y}]}\right)^2 = Var(\tilde{X}) \leq \frac{(b-a)^2}{4}$$

where $\tilde{X}$ is a random variable that satisfies $a \leq \tilde{X} \leq b$. Therefore, we have:

$$\psi(\lambda) \leq \int_0^\lambda \int_0^t \frac{(b-a)^2}{4} dsdt = \frac{(b-a)^2\lambda^2}{8}$$

A mean zero random variable is σ^2-subgaussian if $\psi(\lambda) \leq \frac{\sigma^2\lambda^2}{2}$. Thus, we have:

$$\frac{\sigma^2\lambda^2}{2} = \frac{(b-a)^2\lambda^2}{8}$$
$$\sigma^2 = \frac{(b-a)^2}{4}$$

Thus, $X - E[X]$ is $\frac{(b-a)^2}{4}$-subgaussian.

(b) The constant $\frac{(b-a)^2}{4}$ is the best possible constant. To show this, we use the example of a random variable X that takes on the values a and b with equal probability. Then, we have:

$$E[X] = \frac{a+b}{2}$$
$$Var(X) = E[X^2] - (E[X])^2 = \frac{a^2+b^2}{2} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{4}$$

Let $Y = X - E[X]$. Then, we have:

$$E[Y] = E[X - E[X]] = E[X] - E[X] = 0$$
$$Var(Y) = Var(X) = \frac{(b-a)^2}{4}$$

Now, suppose Y is σ^2-subgaussian. Then

$$\psi_Y(\lambda) = \log E[e^{\lambda Y}] \leq \frac{\sigma^2\lambda^2}{2}$$

Since $\psi_Y''(0) = Var(Y)$ and $\left(\frac{\sigma^2\lambda^2}{2}\right)'' = \sigma^2$, the subgaussian bound near $\lambda = 0$ implies

$$\frac{(b-a)^2}{4} \leq \sigma^2$$

Thus, $Y = X - E[X]$ cannot be σ^2-subgaussian for any $\sigma^2 < \frac{(b-a)^2}{4}$. Therefore, the constant $\frac{(b-a)^2}{4}$ cannot be improved in general.

2

Let X be a random variable where $E[X] = 0$, $Var(X) \leq \sigma^2$, and $|X| \leq b$ almost surely for a finite constant b. For any $\lambda \in (-3/b, 3/b)$, we can show that $\psi(\lambda) \leq \frac{\lambda^2\sigma^2}{2(1-|\lambda|b/3)}$ by using the Taylor exapnsion of $e^{\lambda X}$:

$$e^{\lambda X} = 1 + \lambda X + \frac{\lambda^2 X^2}{2!} + \frac{\lambda^3 X^3}{3!} + \cdots$$
$$E[e^{\lambda X}] = 1 + \frac{\lambda^2 E[X^2]}{2!} + \frac{\lambda^3 E[X^3]}{3!} + \cdots$$

We can define an upper bound $\lambda^n X^n \leq |\lambda|^n |X|^n$, but since $|X| \leq b$, we have $|X|^n \leq b^{n-2}|X|^2$. So:

$$E[e^{\lambda X}] \leq 1 + E[X^2]\left(\frac{\lambda^2}{2!} + \frac{|\lambda|^3 b}{3!} + \frac{|\lambda|^4 b^2}{4!} + \cdots\right)$$
$$E[X] = 0, \quad E[X^2] = Var(X) \leq \sigma^2 \therefore$$
$$E[e^{\lambda X}] \leq 1 + \sigma^2\left(\frac{\lambda^2}{2!} + \frac{|\lambda|^3 b}{3!} + \frac{|\lambda|^4 b^2}{4!} + \cdots\right)$$
$$E[e^{\lambda X}] \leq 1 + \sigma^2\left(\frac{\lambda^2}{2}\right)\left(1 + \frac{|\lambda| b}{3} + \frac{(|\lambda| b)^2}{12} + \cdots\right)$$

But we can recognize the series as a geometric series for the third term onward since $\frac{|\lambda|^4 b^2}{4!} \leq \frac{\lambda^2 b^2}{2}\left(\frac{|\lambda| b}{3}\right)^2$. Thus, we can write:

$$E[e^{\lambda X}] \leq 1 + \frac{\lambda^2 \sigma^2}{2}\left(1 + \frac{|\lambda| b}{3} + \left(\frac{|\lambda| b}{3}\right)^2 + \cdots\right)$$

We know that $|\lambda| < \frac{3}{b}$, so $\frac{|\lambda| b}{3} < 1$ and we can sum the geometric series using $r = \frac{|\lambda| b}{3}$:

$$E[e^{\lambda X}] \leq 1 + \frac{\lambda^2 \sigma^2}{2}\left(\frac{1}{1 - |\lambda| b/3}\right)$$
$$E[e^{\lambda X}] \leq 1 + \frac{\lambda^2 \sigma^2}{2(1 - |\lambda| b/3)}$$
$$log(1 + x) \leq x \therefore$$
$$\log E[e^{\lambda X}] = \psi(\lambda) \leq \frac{\lambda^2 \sigma^2}{2(1 - |\lambda| b/3)}$$

3

(a) Since X is a mean-zero random variable, and $X_+ = \max(X, 0)$, for any $t > 0$:

$$P[X \geq t] = P[X_+ \geq t]$$
$$P[X_+ \geq t] = P[X_+^p \geq t^p]$$
$$\text{By Markov's inequality, } P[X_+^p \geq t^p] \leq \frac{E[X_+^p]}{t^p}$$
$$P[X_+ \geq t] \leq \inf_{p>0} \frac{E[X_+^p]}{t^p}$$

For the second inequality, using $p \to 0_+$, we get that for each fixed x:

$$x_+^p \to \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases} \therefore$$

$$X_+^p \to \mathbf{1}_{X>0} \therefore$$

$$\frac{E[X_+^p]}{t^p} \to \frac{P[X > 0]}{1} = P[X > 0]$$

Let $p = \lambda t$ for $\lambda > 0$. Then, since $t > 0$, we have $p > 0$, thus:

$$\inf_{p>0} \frac{E[X_+^p]}{t^p} \leq \frac{E[X_+^{\lambda t}]}{t^{\lambda t}}$$

$$\frac{x_+^{\lambda t}}{t^{\lambda t}} = \left(\frac{x}{t}\right)^{\lambda t} = e^{\lambda t \log(x/t)}$$

$$\log\left(\frac{x}{t}\right) \leq \frac{x}{t} - 1$$

$$\lambda t \log\left(\frac{x}{t}\right) \leq \lambda(x - t)$$

$$\left(\frac{x}{t}\right)^{\lambda t} \leq e^{\lambda(x-t)} \therefore$$

$$\frac{x_+^{\lambda t}}{t^{\lambda t}} \leq e^{\lambda(x-t)} \therefore$$

$$\frac{E[X_+^{\lambda t}]}{t^{\lambda t}} \leq E[e^{\lambda(X-t)}] = e^{-\lambda t} E[e^{\lambda X}] \therefore$$

$$\inf_{p>0} \frac{E[X_+^p]}{t^p} \leq e^{-\lambda t} E[e^{\lambda X}] \quad \forall \lambda > 0$$

For $\lambda = 0$, we have from the hint $e^{-\lambda t} E[e^{\lambda X}] = 1 \geq P[X > 0]$. Since the inequality holds over all $\lambda \geq 0$, we can take the infimum over $\lambda \geq 0$:

$$\inf_{p>0} \frac{E[X_+^p]}{t^p} \leq \inf_{\lambda \geq 0} e^{-\lambda t} E[e^{\lambda X}]$$

which gives us the full inequality:

$$P[X \geq t] \leq \inf_{p>0} \frac{E[X_+^p]}{t^p} \leq \inf_{\lambda \geq 0} e^{-\lambda t} E[e^{\lambda X}]$$

(b) Let $X = \epsilon Y$ where ϵ is a Rademacher random variable independent of Y, where $P[Y \geq y] = y^{-\alpha}$ for $y \geq 1$ and $\alpha > 1$. For $t \geq e$, the exact tail probability $P[X \geq t]$ is:

$$P[X \geq t] = P[\epsilon Y \geq t] = P[\epsilon = 1, Y \geq t] + P[\epsilon = -1, Y \geq t]$$

$$P[X \geq t] = \frac{1}{2} P[Y \geq t] + 0 = \frac{1}{2} t^{-\alpha}$$

For the first infimum, the exact tail probability $P[X \geq t]$ is:

$$X_+ = \begin{cases} Y, & \epsilon = 1 \\ 0, & \epsilon = -1 \end{cases} \therefore$$

$$E[X_+^p] = \frac{1}{2}E[Y^p] + \frac{1}{2}E[0^p] = \frac{1}{2}E[Y^p]$$

$$E[Y^p] = \int_1^\infty y^p \alpha y^{-\alpha-1} dy = \frac{\alpha}{\alpha - p} \quad \text{for } 0 < p < \alpha \therefore$$

$$\inf_{p>0} \frac{E[X_+^p]}{t^p} = \frac{\alpha}{2(\alpha - p)t^p}$$

We need to minimize this:

$$\log\left(\frac{\alpha}{2(\alpha - p)t^p}\right) = \log\left(\frac{\alpha}{2}\right) - \log(\alpha - p) - p\log(t)$$

$$\frac{d}{dp}\left(\log\left(\frac{\alpha}{2}\right) - \log(\alpha - p) - p\log(t)\right) = 0$$

$$\frac{1}{\alpha - p} - \log(t) = 0 \therefore$$

$$p = \alpha - \frac{1}{\log(t)} \therefore$$

$$\inf_{p>0} \frac{E[X_+^p]}{t^p} = \frac{\alpha}{2(1/\log(t))t^{\alpha - 1/\log(t)}} = \frac{\alpha \log(t)}{2} t^{-\alpha + 1/\log(t)} = \frac{e\alpha \log(t)}{2et^\alpha}$$

Compared to the original tail probability $P[X \geq t] = \frac{1}{2}t^{-\alpha}$, the overestimate factor is:

$$\frac{e\alpha \log(t)}{2et^\alpha} \cdot \frac{2t^\alpha}{1} = e\alpha \log(t)$$

For the second infimum, we have:

$$E[e^{\lambda X}] = \frac{1}{2}E[e^{\lambda Y}] + \frac{1}{2}E[e^{-\lambda Y}]$$

But $E[e^{\lambda Y}] = \int_1^\infty e^{\lambda y} \alpha y^{-\alpha-1} dy$ diverges for any $\lambda > 0$, so we can only consider $\lambda = 0$. Thus, the second infimum is:

$$\inf_{\lambda \geq 0} e^{-\lambda t} E[e^{\lambda X}] = 1$$

Compared to the original tail probability $P[X \geq t] = \frac{1}{2}t^{-\alpha}$, the overestimate factor is:

$$1 \cdot \frac{2t^\alpha}{1} = 2t^\alpha$$

4

(a) If $X = 0$, then for $E[e^{X^2/K^2}] = E[e^0] = 1 \leq 2$, therefore $||X||_{\psi_2} = \inf\{K > 0 : E[e^{X^2/K^2}] \leq 2\} = 0$. Suppose $X \neq 0$, then for $K \to 0^+$, $e^{X^2/K^2} \to \infty$ and consequently $E[e^{X^2/K^2}] \to \infty > 2$. Thus, $||X||_{\psi_2} > 0$, which contradicts $||X||_{\psi_2} = 0$. Therefore, $||X||_{\psi_2} = 0$ if and only if $X = 0$.

(b) $||cX||_{\psi_2} = |c|||X||_{\psi_2}$. If $c = 0$, both sides are zero. If $c \neq 0$, then:

$$||cX||_{\psi_2} = \inf\{K > 0 : E[e^{(cX)^2/K^2}] \leq 2\}$$
$$||cX||_{\psi_2} = \inf\{K > 0 : E[e^{X^2/(K/|c|)^2}] \leq 2\}, \quad K' = K/|c|$$
$$||cX||_{\psi_2} = |c| \inf\{K' > 0 : E[e^{X^2/(K')^2}] \leq 2\} = |c|||X||_{\psi_2}$$

(c) Let $a > ||X||_{\psi_2}$ and $b > ||Y||_{\psi_2}$. Then, we have:

$$E[e^{X^2/a^2}] \leq 2, \quad E[e^{Y^2/b^2}] \leq 2$$
$$E[e^{(X+Y)^2/(a+b)^2}] = E[e^{((a/(a+b))(X/a)+(b/(a+b))(Y/b))^2}]$$
$$\text{Let } \theta = \frac{a}{a+b}, \quad 1-\theta = \frac{b}{a+b}$$
$$(\theta(X/a) + (1-\theta)(Y/b))^2 \leq (\theta)(X/a)^2 + (1-\theta)(Y/b)^2$$
$$E[e^{(X+Y)^2/(a+b)^2}] \leq E[e^{(\theta)(X/a)^2+(1-\theta)(Y/b)^2}]$$
$$E[e^{(X+Y)^2/(a+b)^2}] \leq E[e^{(\theta))(X/a)^2}e^{(1-\theta)(Y/b)^2}]$$
$$E[e^{(X+Y)^2/(a+b)^2}] \leq E[e^{(\theta)(X/a)^2}]E[e^{(1-\theta)(Y/b)^2}]$$
$$E[e^{(X+Y)^2/(a+b)^2}] \leq (E[e^{(X/a)^2}])^{\theta}(E[e^{(Y/b)^2}])^{1-\theta}$$
$$E[e^{(X+Y)^2/(a+b)^2}] \leq (E[e^{(X/a)^2}]E[e^{(Y/b)^2}])^{\theta+(1-\theta))}$$
$$E[e^{(X+Y)^2/(a+b)^2}] \leq 2^1 = 2, \therefore$$
$$||X+Y||_{\psi_2} \leq a + b = ||X||_{\psi_2} + ||Y||_{\psi_2}$$

5

(a) Let X be a random variable where $E[X] = 0$, $Var[X] = \sigma^2$, and $X \leq b$ with probability 1. Then, we can show that for all $\lambda \geq 0$:

$$\log E[e^{\lambda X}] \leq \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}$$

by starting with $e^{\lambda X} = 1 + \lambda X + (e^{\lambda X} - 1 - \lambda X)$ and bounding the last term. Because $X \leq b$ and $\lambda \geq 0$, $\lambda X \leq \lambda b$. From the hint, we can then

know that:

$$\frac{e^{\lambda X} - 1 - \lambda X}{(\lambda X)^2} \leq \frac{e^{\lambda b} - 1 - \lambda b}{(\lambda b)^2}$$
$$e^{\lambda X} - 1 - \lambda X \leq \frac{X^2(e^{\lambda b} - 1 - \lambda b)}{b^2}$$
$$e^{\lambda X} \leq 1 + \lambda X + \frac{X^2(e^{\lambda b} - 1 - \lambda b)}{b^2}$$
$$E[e^{\lambda X}] \leq 1 + \lambda E[X] + \frac{E[X^2](e^{\lambda b} - 1 - \lambda b)}{b^2}$$
$$E[X] = 0, \quad E[X^2] = Var[X] = \sigma^2 \therefore$$
$$E[e^{\lambda X}] \leq 1 + \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}$$
$$\log(E[e^{\lambda X}]) \leq \log\left(1 + \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right)$$
$$\log\left(1 + \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right) \leq \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \therefore$$
$$\log(E[e^{\lambda X}]) \leq \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}$$

(b) Now, with $X_1, \ldots, X_n$ independent random variables equal in law to X, for all $t \geq 0$, we can show that:

$$P\left[\sum_{i=1}^{n} X_i \geq t\right] \leq \exp\left(-\frac{n\sigma^2}{b^2} h\left(\frac{bt}{n\sigma^2}\right)\right)$$

where $h(x) = (1+x)\log(1+x) - x$ by using the Chernoff bound. From part (a) and because each X_i is independent and equal in law to X, we have:

$$E[e^{\lambda \sum_{i=1}^{n} X_i}] = E\left[\prod_{i=1}^{n} e^{\lambda X_i}\right] = (E[e^{\lambda X}])^n$$
$$\log E[e^{\lambda \sum_{i=1}^{n} X_i}] = n \log E[e^{\lambda X}] \leq \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} \therefore$$
$$E[e^{\lambda \sum_{i=1}^{n} X_i}] \leq \exp\left(\frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right)$$

Now using the Chernoff bound, we have:

$$P[\sum_{i=1}^{n} X_i \geq t] \leq \exp\left(-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right)$$

Minimizing the right-hand side over $\lambda \geq 0$, we have:

$$\frac{d}{d\lambda}\left(-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}\right) = 0$$
$$-t + \frac{n\sigma^2(e^{\lambda b} b - b)}{b^2} = 0 \therefore$$
$$e^{\lambda b} = 1 + \frac{bt}{n\sigma^2} \therefore$$
$$\lambda^* = \frac{1}{b}\log\left(1 + \frac{bt}{n\sigma^2}\right)$$

Let $x = \frac{bt}{n\sigma^2}$. Then, we have:

$$\lambda^* = \log(1+x), \quad e^{\lambda^* b} = 1 + x, \quad t = \frac{n\sigma^2 x}{b}$$
$$\therefore$$
$$-\lambda^* t = -\frac{n\sigma^2 x \log(1+x)}{b^2}$$
$$\frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} = \frac{n\sigma^2((1+x) - 1 - \log(1+x))}{b^2} = \frac{n\sigma^2(x - \log(1+x))}{b^2}$$
$$\therefore$$

$$-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} = -\frac{n\sigma^2 x \log(1+x)}{b^2} + \frac{n\sigma^2(x - \log(1+x))}{b^2}$$
$$-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} = -\frac{n\sigma^2}{b^2}(x\log(1+x) + x - \log(1+x))$$
$$-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} = -\frac{n\sigma^2}{b^2}((1+x)\log(1+x) - x)$$
$$-\lambda t + \frac{n\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2} = -\frac{n\sigma^2}{b^2}h(x)$$

Then finally, substituting back $x = \frac{bt}{n\sigma^2}$, we have:

$$P\left[\sum_{i=1}^{n} X_i \geq t\right] \leq \exp\left(-\frac{n\sigma^2}{b^2}h\left(\frac{bt}{n\sigma^2}\right)\right)$$

6

(a) We can show that if $X_1, \ldots, X_n$ are mean-zero and each X_i is σ_i^2-subgaussian, then without any assumption on their dependence,

$\sum_{i=1}^n X_i$ is $\sum_{i=1}^n \sigma_i^2$-subgaussian. Let $A = \sum_{i=1}^n \sigma_i$ and weights $w_i = \frac{\sigma_i}{A}$. Then, we have:

$$\sum_{i=1}^n w_i = 1, \quad w_i \geq 0$$

$$w_i \frac{X_i}{\sigma_i} = \frac{X_i}{A} \therefore$$

$$\sum_{i=1}^n w_i \frac{X_i}{\sigma_i} = \sum_{i=1}^n \frac{X_i}{A}$$

$$\sum_{i=1}^n X_i = A \sum_{i=1}^n w_i \frac{X_i}{\sigma_i} \therefore$$

Because $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$, we can use Jensen's inequality and the convexity of e^x to show that:

$$e^{\lambda A \sum_{i=1}^n w_i (X_i/\sigma_i)} \leq \sum_{i=1}^n w_i e^{\lambda A (X_i/\sigma_i)}$$

$$E[e^{\lambda A \sum_{i=1}^n X_i}] \leq E\left[\sum_{i=1}^n w_i e^{\lambda A (X_i/\sigma_i)}\right] = \sum_{i=1}^n w_i E[e^{\lambda A (X_i/\sigma_i)}]$$

Since X_i is σ_i^2-subgaussian, we have:

$$E[e^{tX_i}] \leq e^{t^2\sigma_i^2/2}, \text{ if } t = \lambda A/\sigma_i$$

$$E[e^{\lambda A (X_i/\sigma_i)}] \leq e^{(\lambda A/\sigma_i)^2 \sigma_i^2/2} = e^{\lambda^2 A^2/2}$$

$$E[e^{\lambda A \sum_{i=1}^n X_i}] \leq \sum_{i=1}^n w_i e^{\lambda^2 A^2/2} = e^{\lambda^2 A^2/2}$$

Recall $A = \sum_{i=1}^n \sigma_i$, so we have:

$$E[e^{\lambda \sum_{i=1}^n X_i}] \leq \exp\left(\frac{\lambda^2}{2}\left(\sum_{i=1}^n \sigma_i\right)^2\right)$$

thus $\sum_{i=1}^n X_i$ is $\left(\sum_{i=1}^n \sigma_i\right)^2$-subgaussian. This bound can not be improved in general. Let ϵ be a Rademacher random variable and let $X_i = \sigma_i \epsilon$. Then, we have:

$$\sum_{i=1}^n X_i = \sum_{i=1}^n \sigma_i \epsilon = \left(\sum_{i=1}^n \sigma_i\right) \epsilon \therefore$$

$$Var(S) = \left(\sum_{i=1}^n \sigma_i\right)^2$$

Any mean-zero random variable is σ^2-subgaussian if $\sigma^2 \geq Var(X)$, so $\sum_{i=1}^n X_i$ is $(\sum_{i=1}^n \sigma_i)^2$-subgaussian, and the bound cannot be improved in general.

(b) Let $k \geq 2$ and $\epsilon_1, \ldots, \epsilon_k$ be independent Rademacher random variables. For every nonempty subset $S \subseteq \{1, \ldots, k\}$, let $X_S = \prod_{i \in S} \epsilon_i$, giving $n = 2^k - 1$ random variables and their respective sum $W = \sum_S X_S$.

i. We can show each X_S is a Rademacher random variable. Picking any $j \in S$, $X_S = \epsilon_j \prod_{i \in S, i \neq j} \epsilon_i$. Since ϵ_j is a Rademacher variable, by conditioning on the other ϵ_i's, $\prod_{i \in S, i \neq j} \epsilon_i$ is a constant, and thus X_S is a Rademacher variable. X_S and X_T are independent for any $S \neq T$:

$$X_S X_T = \left(\prod_{i \in S} \epsilon_i\right)\left(\prod_{j \in T} \epsilon_j\right)$$
$$X_S X_T = \prod_{i \in S \triangle T} \epsilon_i$$

Because $S \neq T$, $S \triangle T$ is nonempty, and thus $X_S X_T$ is a Rademacher variable with expectation 0 from the prior proof. Thus, $Cov(X_S, X_T) = E[X_S X_T] - E[X_S]E[X_T] = 0 - 0 = 0$, and X_S and X_T are uncorrelated, proving their independence.

ii. Recognize:

$$\prod_{i=1}^k (1 + \epsilon_i) = 1 + \sum_{S \subseteq \{1,\ldots,k\}} X_S = 1 + W \therefore$$
$$W = \prod_{i=1}^k (1 + \epsilon_i) - 1$$

Because each ϵ_i is either -1 or 1, a single $\epsilon_i = -1$ will make the product 0, making $W = -1$. Only if all $\epsilon_i = 1$ will the product be nonzero, making $W = 2^k - 1 = n$. Since all ϵ_i are independent, $P(\epsilon_1 = 1, \ldots, \epsilon_k = 1) = P(W = n) = (1/2)^k$, thus $P(W = n) = 1/(n+1)$. n is also the largest possible value of W, so $P(W \geq n) = P(W = n) = 1/(n+1)$.

iii. If W were σ^2-subgaussian, then we would have:

$$P(W \geq t) \leq e^{-t^2/(2\sigma^2)}$$
$$P(W \geq n) \leq e^{-n^2/(2\sigma^2)} \therefore$$
$$\frac{1}{n+1} \leq e^{-n^2/(2\sigma^2)}$$
$$-\log(n+1) \leq -\frac{n^2}{2\sigma^2}$$
$$\log(n+1) \geq \frac{n^2}{2\sigma^2}$$
$$\sigma^2 \geq \frac{n^2}{2\log(n+1)}$$

Thus, W cannot be σ^2-subgaussian for any $\sigma^2 < \frac{n^2}{2\log(n+1)}$. Chebyshev's inequality gives us:

$$P(W \geq n) \leq P(|W| \geq n) \leq \frac{Var(W)}{n^2} = \frac{2^k - 1}{n^2} = \frac{n}{n^2} = \frac{1}{n}$$

overestimating the tail probability by a factor of $\frac{n+1}{n}$. Hoeffding's inequality gives us:

$$P(W \geq n) \leq e^{-2n^2/(4n)} = e^{-n/2}$$

which overestimates the tail probability by a factor of $(n+1)e^{-n/2}$. But this is less than 1 for any $n \geq 3$, so it is not a valid upper bound. Pairwise independence is not enough to justify the Hoeffding bound, but gives the correct variance behavior (just not the full subgaussian tail).

AI Usage: All homework problems were written in VSCode and compiled using LaTeX, with assistance from ChatGPT and Github Copilot inline suggestions.